

Operating System (OS) Midterm Exam

Exam time 2 hours (PM 1:30- 3:30)

1. What are the advantages of "microkernels"?

A :

The advantage of microkernel is that the operating system is easy to expand and maintain, if a new function is added to the user layer, the new function can usually be used without modifying the kernel, even if the kernel needs to be modified, there is no need to change too much, because the kernel itself will involve various functions. There are not many parts, so for different hardware architectures, as long as the hardware-related parts are slightly modified, the kernel can be easily ported over. Microkernels also provide a more secure and reliable execution environment because most of the work performed by the user program does not involve the kernel, but only uses the services in the user layer, so if one service fails, it will not affect the others.

2. What is the main job of a CPU? How many tasks can a CPU perform at once?

A :

(1) Specializes in various calculations and is responsible for communicating with peripherals.

(2) Only one

3. What are the steps involved in the execution of the command?

A : Command extraction, command decoding, data extraction, command execution, and data storage.

4. Suppose there is a computer system that does not provide system calls, but it does provide interrupt handling. How to simulate the mechanism of system calls on such a computer system?

A :

You can define an interrupt handling for system calls, and when users want to call system calls, they can use interrupt processing directly.

5. Please explain what status is usually available for itineraries?

A :

Created: The state when the process was first created.

Ready: The itinerary is ready and ready to be executed.

Suspended: The journey is waiting for an event to occur, usually waiting for the I/O operation to complete.

Execution: The process is running, that is, the CPU is being used.

Terminated: The trip has ended, but some information remains in the system.

Displace/Ready: Procedures can be executed, but are currently overridden in secondary memory.

Displace/Suspend: The process is waiting for an event to occur and is overridden into secondary memory.

6. Briefly describe common queues in the operating system and explain their purpose.

A :

Job queue: When a process is allowed to enter the system, it is placed in the job queue.

Ready Queue: When a process is ready to run, it is moved to the ready queue.

Wait Queue: Processes that wait for certain events are moved into a wait queue.

Device Queue: Processes waiting for I/O devices are moved to the device queue.

7. Please explain the relationship between CPU burst, I/O burst and CPU-dominated and I/O main processes.

A :

The period during which the process continues to use the CPU is called a CPU burst, and the period during which it focuses on I/O operations is called an I/O burst.

CPU-based processes usually consist of some longer CPU bursts, while I/O-based processes consist of many short CPU bursts.

8. Describe the difficulty of implementing SJF scheduling.

A :

Because it is difficult to know the exact length of the next CPU burst time in a journey, we can only use certain estimation methods to get an estimate.

9. In the RR scheduling method, what is the impact of time slices that are too long or too short?

A :

If the time slice is too long, the RR scheduling method is like the FCFS scheduling method; On the other hand, if the time slice is too short, it will cause a lot of additional body switching burden.

10. Suppose there are four processes P1, P2, P3, and P4, all of which arrive at time 0, in the order of P1, P2, P3, and P4, and their CPU burst times are as follows :

<u>Itinerary</u>	<u>CPU burst time (ms)</u>	<u>Priority</u>
P ₁	6	2
P ₂	5	0
P ₃	7	3
P ₄	4	1

(15%)

- What is the order in which the processes are executed when using FCFS, SJF, non-preemptive priority (the lower the number, the higher the priority), and RR (time slice = 3) scheduling methods ?
- What is the average waiting time for all process(trips)?
- What is the response time for each process(trip)?

A :

a.

The schedule for FCFS is as follows :

P ₁		P ₂		P ₃		P ₄	
0	6	11	18	22			

The SJF schedule is as follows :

P ₄	P ₂	P ₁	P ₃	
0	4	9	15	22

The schedule for non-preemptive priority is as follows :

P ₂		P ₄	P ₁		P ₃		
0	5	9	15	22			

The schedule for RR is as follows :

P ₁	P ₂	P ₃	P ₄	P ₁	P ₂	P ₃	P ₄	P ₃	
0	3	6	9	12	15	17	20	21	22

b.

Average wait time for FCFS :

$$(0 + 6 + 11 + 18) / 4 = 8.75 \text{ milliseconds}$$

Average wait time for SJF :

$$(0 + 4 + 9 + 15) / 4 = 7 \text{ milliseconds}$$

The average wait time for non-preemptive priority :

$$(0 + 5 + 9 + 15) / 4 = 7.25 \text{ milliseconds}$$

Average wait time for RR :

$$((12-3) + (15-3) + (20-3) + (21-6)) / 4 = 13.25 \text{ milliseconds}$$

c.

	FCFS	SJF	Priority	RR
P ₁	6	15	15	15
P ₂	11	9	5	17
P ₃	18	22	22	22
P ₄	22	4	9	21